25. Volume and Monte Carlo

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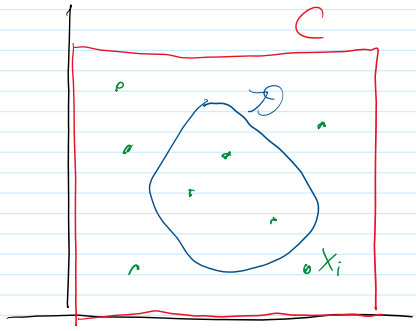
Last time: Region $\mathcal{D} \subset C = [0,1]^{\times d}$



Do Uniform Monte Carlo Sampling

$X_1, \dots, X_n \in \mathbb{R}^d$ iid $\sim$ Unif(C)

each entry $X_{ij} \overset{iid}{\sim}$ Unif$[0,1]$

Note: $\mathrm{Vol}(C) = 1$

$$\mathbb{P}(X \in \mathcal{D}) = \frac{\mathrm{Vol}(\mathcal{D})}{\mathrm{Vol}(C)} = \mathrm{Vol}(\mathcal{D}) := |\mathcal{D}|$$

Estimate: $\dfrac{|\{X_i \in \mathcal{D}\}|}{n} = \dfrac{\#\text{ pts in } \mathcal{D}}{\text{Tot } \#\text{ pts}}$

We have a random estimate of $\mathrm{Vol}(\mathcal{D})$

$\Rightarrow$ treat it as random and set error bounds

Event $\{X \in \mathcal{D}\}$ is a Bernoulli random variable

$b =$ with probability $|\mathcal{D}|$ of it occuring

- $\mathbb{E}(b) = |\mathcal{D}|$ $\leftarrow$ Unbiased
- $\mathrm{Var}(b) = |\mathcal{D}|(1 - |\mathcal{D}|)$

$$\mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n}\{X_i \in \mathcal{D}\}\right) = \frac{|\mathcal{D}|(1-|\mathcal{D}|)}{n}$$

Variance decreases like $1/n$

Binomial r.v.

What is a good error bound on S/n, $S = \sum_{i=1}^{n}\{X_i \in \mathcal{D}\}$

- Use the Central Limit Theorem

  i.e. Binomial $\approx$ Normal for n large and p not too close to 0 or 1 ($p = |\mathcal{D}|$)

  $\hookrightarrow$ Create a $1-\alpha$ confidence interval using the t-distribution

  $$\left| |\mathcal{D}| - S/n \right| \le t_{\alpha} \cdot \hat{\sigma}$$

use t-distribution

$$\left| \underbrace{|\mathcal{D}|}_{\text{true Vol}} - \underbrace{S/n}_{\text{Est Vol}} \right| \le \underbrace{t_{1-\alpha/2,\, n-1}}_{\text{t-value}} \underbrace{\hat{\sigma}}_{\text{Standard Error}}$$

Problem: We didn't independently estimate the mean S/n and the Std err $\hat{\sigma}$ so this may be slightly wrong.

We could instead construct "Non-Asymptotic" conf. ints or error bounds
(popular in Modern Stats + ML as they are valid for any choice of n)

One of the Most General Results

Chebyshev's Inequality:

Let $Z \in \mathbb{R}$ with zero Mean and $\mathbb{E} Z^2 < \infty$

Then $P(|Z| > \varepsilon) \le \frac{\mathbb{E} Z^2}{\varepsilon^2}$

Apply to the above problem, we set

$Z := S/n - |\mathcal{D}|$ (error in estimation)

$\mathbb{E} Z^2 = \frac{1}{n} |\mathcal{D}| (1 - |\mathcal{D}|)$

Don't Know $|\mathcal{D}|$

$$\therefore \mathbb{P}\left(\left| \frac{S}{n} - |\mathcal{D}| \right| > \varepsilon \right) \le \frac{|\mathcal{D}|(1-|\mathcal{D}|)}{n \varepsilon^2} \le \frac{1}{4 n \varepsilon^2}$$

$P(\text{Error} > \varepsilon) \le \frac{1}{4 n \varepsilon^2} = \alpha$

$$\Rightarrow \varepsilon = \frac{1}{\sqrt{4 n \alpha}}$$

$\therefore$ Chebyshev Interval, $\left| S/n - |\mathcal{D}| \right| \le \frac{1}{\sqrt{4 n \alpha}}$

with probability $\ge 1 - \alpha$

our interval shrinks like $O(n^{-1/2})$

Note: this $\frac{1}{\sqrt{\ }}$ rate is the same as the CLT

Note: this $\frac{1}{\sqrt{n}}$ rate is the same as the CLT but with worse (more general) constants

We have a less general but stronger (sharper) inequality for Bernoulli r.v.s.

Hoeffding's Inequality

Let $B_1, \dots, B_n \overset{iid}{\sim} \text{Bernoulli}(p)$ and $\bar{B} = \frac{1}{n}\sum_{i=1}^{n} B_i$

Then $\mathbb{P}\left(|\bar{B} - p| > \varepsilon\right) \leq 2\exp\left(-2n\varepsilon^2\right)$

exponential in ε rather than $1/\varepsilon^2$